Louis Bensard

Co-founder & CTO · Machine Learning / AI Engineer
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SUMMARY

Co-founder & CTO and hands-on machine learning engineer with 7+ years taking ML systems from research to production. Designed and trained transformer-based deep learning models in PyTorch, built end-to-end MLOps (GPU training, evaluation, inference) and data platforms on AWS, and built and led data science and engineering teams. Grounded in mathematics and statistics, with depth in time-series forecasting, Bayesian modeling, recommender systems and LLM fine-tuning.

EXPERIENCE

SIGHTLINE New York, NY
Co-founder & CTO (previously Founding Head of AI & ML) *Aug 2025 – Present*

- **AI/ML strategy & model architecture:** Own AI/ML strategy end to end. Designed and trained the production forecasting engine from scratch: a custom transformer architecture (PatchTST) in PyTorch with hurdle modeling for intermittent demand, across thousands of item-level time series, reaching sub-5% wMAPE.
- **Training, evaluation & MLOps:** Built the full ML lifecycle in-house: feature engineering, hyperparameter optimization, multi-seed ensembling, backtesting, GPU model training on AWS Batch, experiment tracking in Weights & Biases, and training and inference pipelines in production.
- **Decision intelligence & recommendations:** Architected the recommendation layer that converts model output into ranked, explainable next-best actions for operators, using composite impact scoring.
- **Data platform & cloud infrastructure:** Designed and built the data platform behind all ML: multi-source ETL pipelines with data-quality validation and idempotent bulk loading into PostgreSQL, processing millions of rows on AWS (S3, Lambda, SQS, ECS).
- **Product engineering:** Own the technical architecture of the multi-tenant product (Next.js/TypeScript) and shipped AI features across forecasting, inventory and COGS that enterprise restaurant chains use daily.
- **Technical leadership:** Built the engineering team and department from scratch: hiring process and bar, engineering standards (testing, CI/CD, documentation) and the data and AI roadmap.

Tech: Python, PyTorch, scikit-learn, SQL, PostgreSQL, AWS (Batch, Lambda, S3, SQS, ECS), Docker, Weights & Biases, Next.js, TypeScript, CI/CD

EKIMETRICS New York, NY
Manager & Tech Lead, Data Science *Jul 2024 – Aug 2025*

- **Bayesian modeling:** Built Bayesian marketing-mix models in PyMC with MCMC inference, estimating channel-level effects and ROI from time-series KPI data for Fortune 500 retail and beauty clients.
- **Forecasting & deployment:** Built time-series forecasting models (PyMC, scikit-learn, XGBoost) and took them from scoping through deployment, driving $20M+ in measured client impact.
- **Frameworks & pipelines:** Built reusable modeling frameworks and data pipelines in Python and SQL, used across client projects.
- **Leadership:** Led a team of 9+ data scientists across projects; primary technical contact for 10+ strategic accounts, presenting results to C-level executives.

Tech: Python, PyMC, scikit-learn, XGBoost, pandas, SQL, Bayesian statistics, MCMC

STRATOS AVIATION New York, NY
Founder & Lead ML/AI Engineer *Aug 2023 – Dec 2024*
A "Strava for pilots" iOS app, live on the App Store. Designed, built and shipped solo, from concept to launch in 8 months.

- **Recommender system:** Built personalized airport and route recommendations from nearest-neighbor retrieval over flight time-series and geospatial features, with on-device inference via Core ML.
- **Geospatial ML:** Built geospatial recommendations across 100K airports and 10M+ points of interest.
- **LLM fine-tuning:** Curated the training data and fine-tuned an OpenAI model into a pilot assistant that gives flight performance recommendations, deployed in-app and as an iMessage extension.
- **ML pipelines:** Automated preprocessing, feature selection (Boruta, RFE), model evaluation and Core ML export of scikit-learn models for production.
- **Security:** Built end-to-end encrypted messaging between pilots (RSA-2048 key exchange, AES-256).

Tech: Python, PyTorch, TensorFlow, scikit-learn, Core ML, OpenAI API, Swift, GCP, REST APIs

IPSOS MMA New York, NY
Manager & Tech Lead, Data Science *Jul 2019 – Jun 2023*
Promoted three times, from Senior Analyst (2019) to Manager & Tech Lead (2022).

- **Bayesian modeling:** Built Bayesian and hierarchical marketing-mix models (PyMC3, MCMC inference) on time-series KPI data for Fortune 500 clients.
- **Predictive modeling at scale:** Developed and deployed hundreds of predictive models, from gradient boosting to neural networks (XGBoost, TensorFlow, scikit-learn), on 10B+ data points spanning marketing, macroeconomic and business data.
- **Feature engineering & ML automation:** Improved accuracy with feature engineering and missing-data imputation; automated the team's pipelines for feature selection, Bayesian hyperparameter optimization and evaluation.
- **Interpretability:** Explained model behavior to C-suite executives with SHAP, LIME and interpretable models, on projects worth $4–6M a year.

Tech: Python, PyMC3, TensorFlow, XGBoost, scikit-learn, SQL, AWS, MCMC, SHAP

PANASONIC AVIONICS CORPORATION Fullerton, CA
Machine Learning Engineer *Jan 2019 – May 2019*

- **Recommendation engine:** Built production ML models (scikit-learn, TensorFlow) for an in-flight entertainment recommendation engine, optimizing airlines' movie selection from passenger viewing history.
- **Modeling:** Predicted passenger viewing preferences with 85%+ accuracy using gradient boosting on a 10M x 50 dataset, with multivariate analysis (PCA, LDA); led a 7-person team.

EDUCATION

California State University, Fullerton 2017 – 2019
M.S. Statistics. Summa cum laude, 4.0 GPA, top of class; Phi Kappa Phi.
Graduate coursework: Machine Learning, Deep Learning, Reinforcement Learning, Bayesian Analysis, Advanced Probability Theory, Multivariate Analysis.

Sorbonne Université, Paris 2014 – 2017
B.S. Mathematics and B.S. Computer Science (two degrees).

SKILLS

Machine learning & AI: Deep learning, transformers (PatchTST), large language models (LLMs), LLM fine-tuning, generative AI, NLP, time-series forecasting, recommender systems, nearest-neighbor retrieval, Bayesian inference (MCMC), probabilistic modeling, gradient boosting, ensembling, feature engineering, hyperparameter optimization, model evaluation and backtesting, model interpretability (SHAP)

MLOps & infrastructure: GPU training, training and inference pipelines, model deployment, experiment tracking (Weights & Biases), on-device inference (Core ML), ETL and data pipelines, data quality, AWS (Batch, Lambda, S3, SQS, ECS), GCP, Docker, CI/CD, PostgreSQL

Languages & frameworks: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, PyMC, NumPy, pandas, SQL, TypeScript, Next.js, Swift

Mathematics & statistics: Probability theory, linear algebra, Bayesian statistics, optimization, multivariate analysis, statistical modeling

Leadership: Technical strategy and architecture (0-to-1), AI/ML roadmap, engineering team building and hiring, engineering standards, cross-functional leadership, executive communication

ADDITIONAL

Licensed private pilot · Competitive rock climber (France, national level) · English, French (native), Spanish